

NODE Platform

Hardware Reference · 8-Channel Addressable LED Controller Family

Wateefy Electronics

The power section is the product. The module is a consumable.

Document rev: 4.0 · Board rev: V1a / V1b · Date: 2026-09-01

Scope

This document covers the hardware common to every board in the NODE family — power architecture, input protection, output stage, environmental data and build process. It applies to **NODE Mini**, **NODE Flex** and **NODE Backbone** alike.

What differs between variants — the MCU module, the logic rail source, board dimensions, GPIO mapping and per-board accepted limitations — is documented in each board's own **Variant Specification**. Where a variant departs from anything stated here, its supplement says so explicitly in *Deltas from the Platform Reference*.

Read this document for how the board protects your pixels. Read the supplement for which board to buy.

Components are specified by function and rating. Specific manufacturers and part numbers are selected at build time and may change with availability; every substitution is held to the ratings stated here.

1. Power Architecture

Three separate domains, each on its own copper region:

Domain	Boundary	What it carries
Input (unprotected)	Screw terminal → ideal-diode stage	Raw supply. Assumed hostile. Transient clamp sits across this node.
Clean (protected)	Ideal-diode output → main fuse	Reverse-blocked, surge-clamped. Bulk capacitance lives here.
Distribution bus (fused)	Main fuse → 8 channel fuses + logic regulator	Everything downstream taps this bus through its own fuse.

NODE Backbone adds a fourth domain: a network-powered logic rail, on its own ground domain, joined to board ground at one defined point. See that board's variant specification.

2. Input & Protection

Parameter	Value	Notes
Pixel input voltage range	5 - 48 VDC	36 V nominal, 48 V absolute maximum
Input connector	Enclosed 2-pole screw terminal	57 A / 10 mm² rated; 30 A design maximum = 53% of rating
Maximum continuous current	30 A	At standard 1 oz outer copper — no copper upgrade required
Main fuse	30 A mini blade, user-replaceable	Holder rated 30 A / 500 V
Reverse-polarity protection	Ideal-diode controller driving a high-side N-channel pass FET	FET held off under reverse input; no current path to load; non-sacrificial and self-recovering
Pass element	80 V N-channel MOSFET, low milliohm $R_{DS(on)}$	Rated above the input clamp voltage, not below it; current rating several times the 30 A design load
Input transient clamp	Bidirectional TVS, 1500 W, 48 V standoff	Conducts in neither direction at rated input; polarity-independent
Bulk capacitance	2 × 1000 µF, 63 V	On the clean rail, downstream of the ideal diode
Ideal-diode support	100 V charge-pump capacitor, 80 V hold-up capacitor	Per the controller's reference design
Polarity indication	Anti-parallel green/red LED pair	Green = correct, red = reversed; shared limit resistor sized for the full input range

Voltage headroom summary: 36 V nominal, 48 V absolute maximum. Every component in the power path is rated **63 V or above**; every semiconductor is **60 V or above**.
Derating against input voltage. Utilization of each component's own rating. **5 V, 12 V and 24 V are the mainstream pixel voltages** and are shown in bold:

Component	Rating	5 V	12 V	24 V	36 V	48 V
Bulk input capacitors	63 V	8 %	19 %	38 %	57 %	76 %
Channel output caps	80 V	6 %	15 %	30 %	45 %	60 %
Ideal-diode hold-up	80 V	6 %	15 %	30 %	45 %	60 %
HV-plane ceramics	100 V	5 %	12 %	24 %	36 %	48 %
Ideal-diode controller	65 V	8 %	18 %	37 %	55 %	74 %
Pass MOSFET V_{DS}	80 V	6 %	15 %	30 %	45 %	60 %

At the **36 V nominal** rating no part on the plane exceeds 60 % of its own limit. At the **48 V absolute maximum** nothing exceeds 80 %. On the 5 V, 12 V and 24 V strips that carry the overwhelming majority of installations, the entire plane runs under 40 %.

3. Outputs

Parameter	Value	Notes
Channels	8	Independent, individually fused — no shared fuses
Output connector	3-pole pluggable terminal, per channel	V+ / DATA / GND; 12 A rated → 63% at the 7.5 A fitted fuse, 83% at the 10 A maximum
Per-channel fuse	7.5 A mini blade fitted, 10 A maximum	25% of holder rating at 7.5 A, 33% at 10 A; user-replaceable
Level translation	Octal logic buffer, DIP-20 socket	Accepts either of two common octal families; 3.3 V drive against a 2.0 V V_{IH} = 1.3 V margin
Data series resistance	330 Ω , axial through-hole, per channel	Limits fault current back into the buffer
Data ESD protection	Sub-picofarad ESD device, per channel, connector side	0.4–0.55 pF junction capacitance; RC $\approx$ 181 ps — no measurable effect on WS281x bit timing
Data drive current	$\approx$ 4 mA per channel	67% of the $\pm$ 6 mA buffer rating
Buffer supply current	$\approx$ 45 mA total	64% of the 70 mA device maximum
Per-channel decoupling	47 μ F electrolytic 80 V + 100 nF ceramic 100 V	Local to each output connector
Channel activity indication	One LED per channel	Data activity on that channel
Fuse-status indication	One LED per channel, fed from the post-fuse node	Lit = that channel's fuse is intact and the bus is live

4b. DMX-512 / RS-485 Output

Parameter	Value	Notes
Ports	4	RJ45, T568B pairing
Transceivers	One RS-485 transceiver per port	Driver enables tied to a shared enable net; transmit data shared across all four
Receive	Port 1 only	The remaining three ports are transmit-only by design
Port protection	TVS array on the connector-side net, plus series resistance per line	Clamp sits ahead of the series limiting, in that order

4. Environmental & Compliance

Parameter	Value
Operating temperature	−20 °C to +60 °C (<i>design target; not qualification tested</i>)
Substrate rating	UL-recognized, JLC-1 marking
RoHS	Compliant
Workmanship	IPC Class 2
Electrical test	Flying-probe, 100% of boards
Traceability	2D barcode serial number, per board

Mounting & Enclosure

No enclosure is included, and none is planned. This is the norm in this segment — bare board, mounted in whatever the installation already uses: a CG1500, a DIN rail, a weatherproof box, a plywood backer.

What NODE adds instead is **conformal coating, applied at the fabrication facility** with connectors, sockets, fuse clips, and switches masked. Competing boards in this class ship uncoated. A coated board in a vented enclosure tolerates condensation and humidity in a way a bare board does not, which matters for the seasonal outdoor installs this was built for.

Four corner mounting holes are provided. *Hole spacing and a downloadable mount STL: to be published at first article.*

Build & Assurance

- **Fabrication** — 4-layer, 1.6 mm, high- T_g substrate, ENIG finish, epoxy-filled and capped vias, IPC Class 2 workmanship
- **Assembly** — professional SMT assembly, flying-probe electrical test on 100% of boards
- **Traceability** — 2D barcode serial on every board; UL-recognized substrate marking
- **Coating** — conformal coating applied at the fab, with connectors, sockets, fuse clips, and switches masked
- **Design review** — every net verified against manufacturer datasheets; schematics independently reviewed by two automated analysis passes with human verification of every finding

Accepted Limitations

- Stated by design, not as oversights:
- **Not automotive qualified.** Protection targets transients credible in a residential or light-commercial installation. Not tested to ISO 7637 load-dump profiles; not intended for vehicle use.
 - **Worst-case clamp coordination.** At the TVS datasheet maximum clamping voltage under a full rated pulse, the voltage presented to downstream components exceeds their ratings. That condition requires a pulse energy source which does not exist in a mains-fed low-voltage installation, and is accepted as non-credible for the intended deployment. At realistic clamp currents the entire downstream chain retains margin.
 - **Over-voltage beyond specification is not survivable.** A supply above the rated input range will conduct the input TVS and open the main fuse. Sacrificial by design — the board is protected, the TVS and fuse are consumables.

